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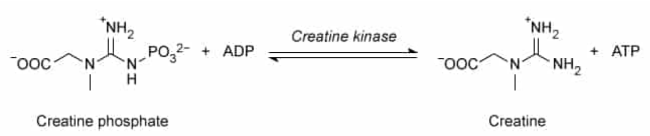

Figure 1 Reaction of creatine phosphate with ADP to produce ATP

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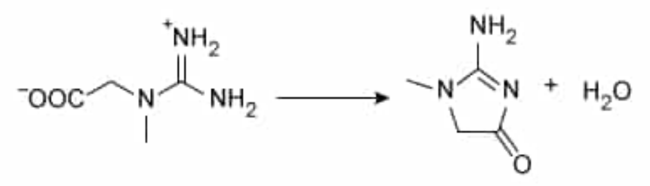


Figure 2 Production of creatinine from creatine

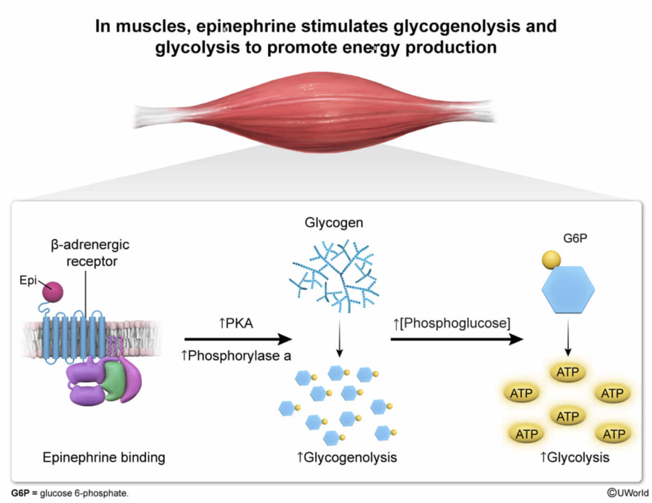
Measurement of creatinine levels in blood serum is a common test to measure a patient's renal function. In patients with healthy kidneys, creatinine is freely filtered at the glomerulus without being reabsorbed later in the nephron; therefore, creatinine levels serve as a useful marker of the glomerular filtration rate. Serum creatinine levels are often measured alongside serum urea levels, also known as blood urea nitrogen (BUN), and the BUN:creatinine ratio assists in diagnosing whether kidney dysfunction is due to intra- or extra-renal causes.

In skeletal muscle cells, epinephrine supports muscular activity by promoting muscular:

- A. glycogenesis and glycolysis. [7%]
- B. glycogenolysis and gluconeogenesis. [28%]
- ✔ C. glycogenolysis and glycolysis. [60%]
- D. glycogenesis and gluconeogenesis. [2%]

Correct
61% answered correctly

Explanation:



Hormones are chemical messengers (eg, peptides, small molecules) that are secreted from a cell and induce a response in a target cell. Hormones can affect multiple pathways within a given cell, as well as multiple organs and tissues within a given organism, to support the current state of the organism.

The question asks about epinephrine's effect on muscle tissue. Epinephrine is a hormone that activates the "**fight-or-flight**" responses of the sympathetic nervous system. For skeletal muscles, these responses include the **mobilization of glycogen** and the **utilization of glucose** to produce energy in the form of ATP.

To mobilize glycogen, epinephrine first binds the **β-adrenergic receptor**, a G protein–coupled receptor (GPCR) that is coupled to a heterotrimeric G protein with a $G_{s\alpha}$ subunit. This initiates a signaling cascade that leads to the production of **3′,5′-cyclic AMP (cAMP)**, activation of **protein kinase A (PKA)**, and the phosphorylation and activation of the enzymes **phosphorylase kinase** and **phosphorylase**. Phosphorylase breaks down the α(1→4) glycosidic bonds of glycogen through phosphorolysis; therefore, **epinephrine stimulates glycogenolysis**.

The product of phosphorylase is glucose 1-phosphate (G1P). G1P is isomerized to glucose 6-phosphate (G6P) by the enzyme phosphoglucomutase. Because **muscle cells do not export glucose or express glucose-6-phosphatase**, the buildup of the substrate G6P therefore leads to an **increase** in glycolysis through Le Châtelier's principle.

Taken together, in skeletal muscle cells, epinephrine leads to an **increase in both glycogenolysis and glycolysis**. These pathways work together to support muscles during the "fight-or-flight" response by providing them with energy in the form of ATP.

(Choice A) Glycogenesis, the increased synthesis of glycogen, is *inhibited* by epinephrine through the PKA-induced phosphorylation of glycogen synthase kinase and glycogen synthase.

(Choice B) Although gluconeogenesis is stimulated by activation of $G_{s\alpha}$-coupled GPCRs in the *liver* (in order for the liver to produce glucose for other tissues), gluconeogenesis is *not* stimulated in skeletal muscle by these hormone-sensitive receptors.

(Choice D) Both glycogenesis and gluconeogenesis are *reduced* in skeletal muscles following epinephrine stimulation.

Educational objective:
Epinephrine activates the β-adrenergic receptor, a $G_{s\alpha}$-coupled G protein–coupled receptor, on skeletal muscle cells. Stimulation of this pathway leads to an increase in glycogenolysis and—in skeletal muscles—an increase in glycolysis, which supports the sympathetic "fight-or-flight" response.

Time spent:0 sec
Subject:
Biochemistry

QID:403998
System:
Metabolic Reactions

During vigorous exercise, epinephrine stimulates β-adrenergic receptors to support increased muscular activity. Despite this, active muscles often still consume more oxygen than can be delivered in time, leading to an oxygen deficit. Under these low-oxygen conditions, muscles utilize anaerobic glycolysis to provide the bulk of their ATP. The NADH that is also produced during glycolysis is reoxidized back to NAD^+ in a process called fermentation, which allows glycolysis to continue.

Aside from glycolysis and fermentation, muscles can also produce additional ATP from a molecule called creatine phosphate (Figure 1). In this way, creatine phosphate serves as an additional source of fuel to anaerobically produce ATP. Once oxygen availability and aerobic respiration is restored (ie, aerobic conditions), creatine phosphate levels are replenished by phosphate transfer from ATP. Both reactions are catalyzed by the enzyme creatine kinase.

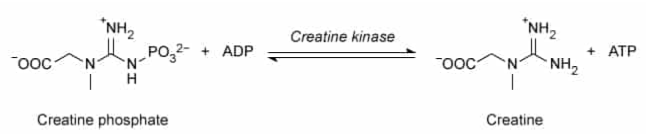


Figure 1 Reaction of creatine phosphate with ADP to produce ATP

In contrast, waste byproducts of creatine are excreted after they are formed by a nonenzymatic, irreversible, intramolecular cyclization reaction converting creatine into the molecule creatinine (Figure 2).

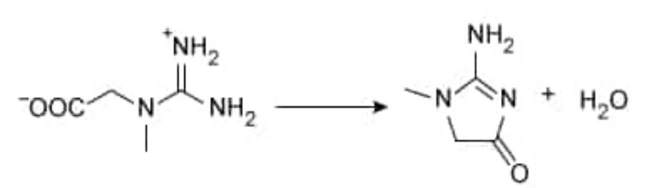


Figure 2 Production of creatinine from creatine

Measurement of creatinine levels in blood serum is a common test to measure a patient's renal function. In patients with healthy kidneys, creatinine is freely filtered at the glomerulus without being reabsorbed later in the nephron; therefore, creatinine levels serve as a useful marker of the glomerular filtration rate. Serum creatinine levels are often measured alongside serum urea levels, also known as blood urea nitrogen (BUN), and the BUN:creatinine ratio assists in diagnosing whether kidney dysfunction is due to intra- or extra-renal causes.

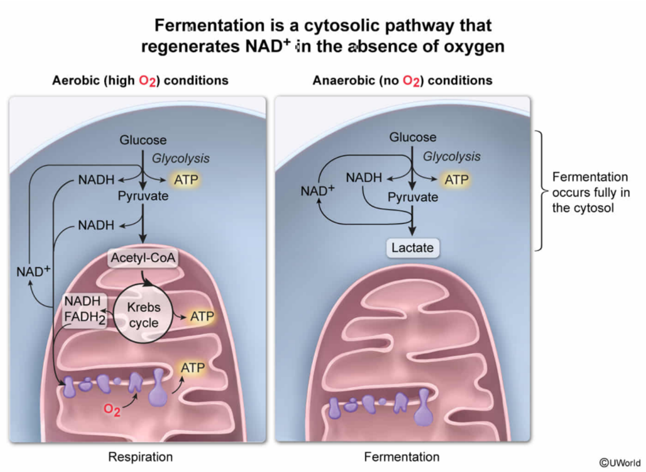
The anaerobic process that replenishes NAD^+ levels occurs in the:

. .

- ◯ A. mitochondrial matrix. [11%]
- ◯ B. lumen of the endoplasmic reticulum. [1%]
- ✔ ◉ C. cytosol. [79%]
- ◯ D. intermembrane space of the mitochondria. [7%]

Correct
78% answered correctly

Explanation:



In eukaryotes, glucose is utilized as a fuel to produce ATP in one of two ways, depending on whether the cell is under aerobic or anaerobic conditions. Both processes begin in the **cytosol** with glycolysis, which consumes one glucose molecule to produce two pyruvate molecules, a net of two ATP molecules, and **two NADH** molecules. Formation of the NADH molecules, which each contain two high-energy electrons, requires the **consumption of two NAD^+** molecules. Therefore, cytosolic **NAD^+ must be regenerated** for glycolysis to continue.

Under aerobic conditions (ie, with oxygen present), either the malate–aspartate shuttle or the glycerol-3-phosphate shuttle are used to oxidize NADH back to NAD^+ in the cytosol, and the high-energy electrons enter the mitochondrial matrix to eventually reduce **oxygen** to water through the electron transport chain. This is cellular respiration.

Under **anaerobic conditions** (ie, without oxygen), there is **no oxygen to receive the high-energy electrons** and therefore NAD^+ cannot be regenerated through respiration. Instead, NAD^+ is anaerobically regenerated through fermentation. Animal cells perform **lactic acid fermentation**, in which the NADH and pyruvate produced during glycolysis react together to **regenerate**

微信Mai_ange

NAD^+ and produce lactate.

The lactic acid is exported to other tissues like the oxygen-rich liver, where it can serve as a substrate for gluconeogenesis (as in the Cori cycle). The regenerated NAD^+ remains in the cell to allow continued anaerobic glycolysis. Because fermentation **does not require** the use of oxygen or the electron transport chain, it can occur even in cells **without mitochondria**, such as red blood cells. Instead, fermentation occurs **completely in the cytosol**.

(Choices A, B, and D) Fermentation occurs completely in the cytosol. It is not compartmentalized into organelles like the mitochondria or endoplasmic reticulum.

Educational objective:
Under anaerobic conditions, glycolysis is followed by fermentation, in which NADH is oxidized back to NAD^+ without requiring oxygen. Fermentation occurs in the cytosol of cells.

Time spent:0 sec
Subject:
Biochemistry

QID:403999
System:
Metabolic Reactions

During vigorous exercise, epinephrine stimulates β-adrenergic receptors to support increased muscular activity. Despite this, active muscles often still consume more oxygen than can be delivered in time, leading to an oxygen deficit. Under these low-oxygen conditions, muscles utilize anaerobic glycolysis to provide the bulk of their ATP. The NADH that is also produced during glycolysis is reoxidized back to NAD^+ in a process called fermentation, which allows glycolysis to continue.

Aside from glycolysis and fermentation, muscles can also produce additional ATP from a molecule called creatine phosphate (Figure 1). In this way, creatine phosphate serves as an additional source of fuel to anaerobically produce ATP. Once oxygen availability and aerobic respiration is restored (ie, aerobic conditions), creatine phosphate levels are replenished by phosphate transfer from ATP. Both reactions are catalyzed by the enzyme creatine kinase.

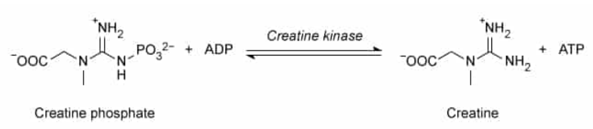


Figure 1 Reaction of creatine phosphate with ADP to produce ATP

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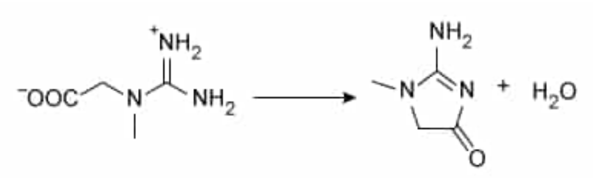

Figure 2 Production of creatinine from creatine

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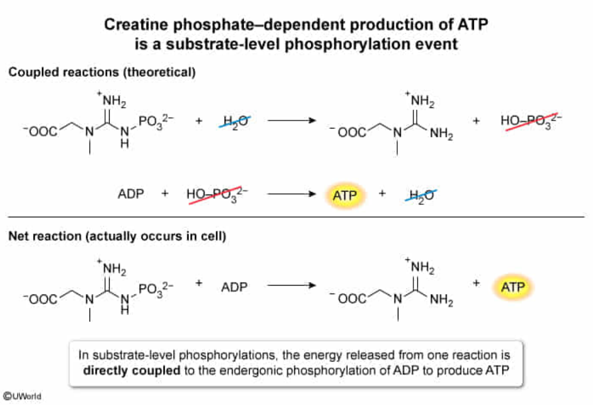
The production of ATP using creatine phosphate is best described as:

- A. substrate-level phosphorylation. [76%]
- B. oxidative phosphorylation. [15%]
- C. photophosphorylation. [1%]

D. phosphorolysis. [7%]

Correct
76% answered correctly

Explanation:



Adenosine triphosphate (ATP) is the major energy currency used by the cell. Because ATP usage is critical for powering many cellular functions, the **phosphorylation of ADP** to form ATP is critical as well. Animal cells tend to create ATP in one of two ways: substrate-level phosphorylation or oxidative phosphorylation.

In substrate-level phosphorylation, the phosphorylation of ADP or another nucleoside diphosphate (NDP) is directly coupled to another reaction, typically the hydrolysis of a high-energy molecule. For example, the two ATP-yielding steps of glycolysis couple ADP phosphorylation to hydrolysis of phosphorylated compounds (ie, 1,3-bisphosphoglycerate, phosphoenolpyruvate). Note that the actual reactions in glycolysis are **transfer** reactions; however, because Gibbs free energy is a **state function**, the sum of the free energies of the theoretical, coupled reactions (ie, phosphorylation of ADP by P_i and hydrolysis of the phosphorylated substrate by water) provide an accurate measure of the free energy of the transfer reaction.

In oxidative phosphorylation, the ADP phosphorylation is **indirectly coupled** to the **oxidation of reduced molecules** (ie, NADH, $FADH_2$). These molecules pass high-energy electrons into an electron transport chain, where the electrons participate in a series of oxidation–reduction reactions that pump protons against their electrochemical gradient before ultimately reducing oxygen to water. The proton gradient forms a proton-motive force that powers the phosphorylation of ADP.

Figure 1 shows that the creatine phosphate–mediated phosphorylation of ADP is a net **phosphoryl-transfer reaction** with no external energy source, implying that ADP phosphorylation is **directly coupled** to **theoretical hydrolysis of creatine phosphate**. Therefore, the creatine phosphate–mediated production of ATP is **substrate-level phosphorylation**.

(Choice B) The creatine phosphate–mediated production of ATP does *not* involve reduced molecules passing electrons to the electron transport chain. Therefore, it is *not* oxidative phosphorylation.

(Choice C) Photophosphorylation occurs in photosynthetic organisms and involves the capture of light energy to produce ATP. The energy for the creatine phosphate–mediated production of ATP comes from chemical energy stored in creatine phosphate, not from light.

(Choice D) Phosphorolysis is the breaking apart of a molecule using inorganic phosphate. In the creatine phosphate–mediated production of ATP, creatine phosphate transfers its phosphate group to ADP; inorganic phosphate does *not* break apart a molecule.

Educational objective:
ATP can be produced by substrate-level phosphorylation or by oxidative phosphorylation. The direct coupling of ADP phosphorylation to the hydrolysis of a high-energy molecule (eg, creatine phosphate) is an example of substrate-level phosphorylation.

Time spent:0 sec	**QID**:404000
Subject:	**System:**
Biochemistry	Metabolic Reactions

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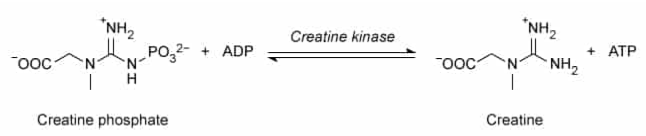

Figure 1 Reaction of creatine phosphate with ADP to produce ATP

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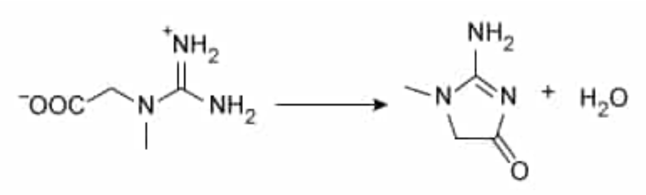

Figure 2 Production of creatinine from creatine

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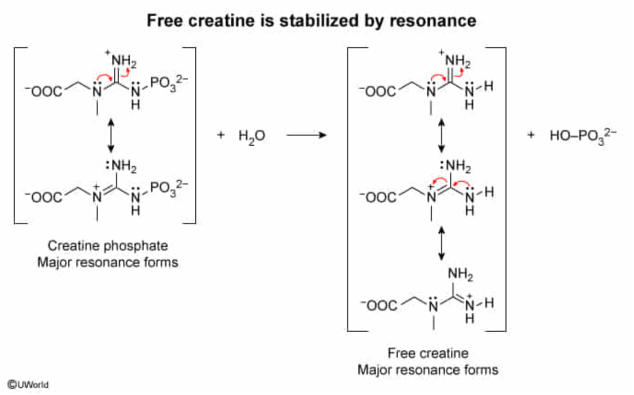
Figure 1 shows that hydrolysis of creatine phosphate can be coupled to phosphorylation of ADP to form ATP. Based on the structures shown in Figure 1, hydrolysis of creatine phosphate most likely provides the energy needed to form ATP because:

I. free creatine has more major resonance forms than creatine phosphate.
II. free creatine undergoes keto–enol tautomerization to become a lower-energy molecule.
III. the reactants of the reaction experience greater charge–charge repulsion than the products.

- A. I only [24%]
- B. I and II only [18%]
- C. II and III only [36%]
- D. I, II, and III [20%]

Correct
25% answered correctly

Explanation:



In biochemistry, adenosine triphosphate (ATP) is considered a **high-energy molecule** because its hydrolysis can be used to power many other reactions. Because free energy is a state function, the hydrolysis of ATP can power nonspontaneous reactions by energetically coupling to them, even if water is not involved in the final net reaction.

Similarly, creatine phosphate and other molecules can also be considered high-energy if their hydrolysis can be coupled to power ATP formation through substrate-level phosphorylation. Chemically, these hydrolysis reactions are highly exergonic, which can occur if the reactants are unstable, the products are stable, or both.

Resonance, the delocalized movement of electron pairs through pi orbitals across at least three adjacent atoms, commonly stabilizes organic and biochemical molecules by decreasing their free energy. The products of creatine phosphate hydrolysis (ie, free creatine and hydrogen phosphate) have many resonance forms that stabilize them.

Furthermore, several of these resonance forms and combinations are less stable or unavailable to unhydrolyzed creatine phosphate; therefore, they are not *major* resonance forms when the phosphate group is present. Taken together, hydrolysis of creatine phosphate produces a molecule with *more major resonance forms*, resulting in a more highly resonance-stabilized structure. This stabilization contributes to a large energy release upon hydrolysis **(Number I)**.

(Number II) Tautomerization is the movement of a proton from one position in a molecule to another, along with a shift in the position of a double bond (eg, keto–enol tautomerization). Tautomerization to form a more stable product contributes to the large negative free energy change of *some* reactions (eg, phosphoenolpyruvate hydrolysis), but free creatine has no ketone or enol groups with which to undergo keto–enol tautomerization.

(Number III) Some reactions, such as ATP hydrolysis, have charge–charge repulsions in the reactants that are not present in the products, so the reaction releases energy as repulsion is relieved. However, creatine phosphate (a reactant) does not experience significant charge–charge repulsion because the negative charges in the molecule are far apart and have a positive charge separating them. In contrast, ATP (a product) has multiple negative charges close together.

Educational objective:
The free energy change upon hydrolysis of high-energy molecules in biochemistry can be explained by several factors, including resonance, tautomerization, and charge–charge repulsion.

Time spent:0 sec
Subject:
Biochemistry

QID:404001
System:
Metabolic Reactions

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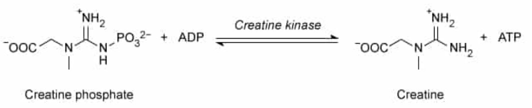


Figure 1 Reaction of creatine phosphate with ADP to produce ATP

In contrast, waste byproducts of creatine are excreted after they are formed by a nonenzymatic, irreversible, intramolecular cyclization reaction converting creatine into the molecule creatinine (Figure 2).

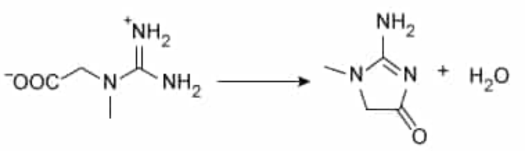


Figure 2 Production of creatinine from creatine

Measurement of creatinine levels in blood serum is a common test to measure a patient's renal function. In patients with healthy kidneys, creatinine is freely filtered at the glomerulus without being reabsorbed later in the nephron; therefore, creatinine levels serve as a useful marker of the glomerular filtration rate. Serum creatinine levels are often measured alongside serum urea levels, also known as blood urea nitrogen (BUN), and the BUN:creatinine ratio assists in diagnosing whether kidney dysfunction is due to intra- or extra-renal causes.

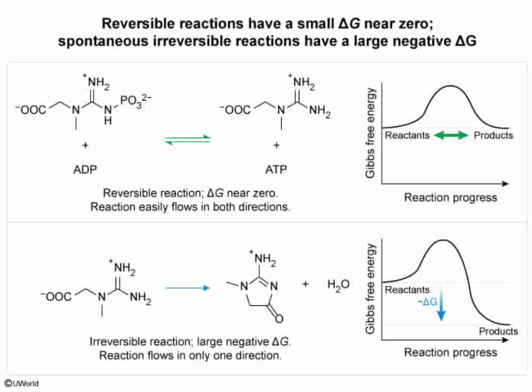
At physiological conditions, the reactions shown in Figures 1 and 2 are best described as having:

- A. a large negative ΔG and a large negative ΔG, respectively. [16%]
- B. a small ΔG near 0 and a large positive ΔG, respectively. [10%]
- C. a large positive ΔG and a large negative ΔG, respectively. [16%]
- ✔ D. a small ΔG near 0 and a large negative ΔG, respectively. [57%]

Correct

57% answered correctly

Explanation:



The change in **Gibbs free energy (∆*G*)** provides important information about a reaction's **spontaneity** (ie, whether it can proceed in a given direction) and its **reversibility**. Reactions with **smaller ∆*G*values near 0** are said to be reversible. Because ∆*G* varies with the concentrations of products and reactants, the spontaneity and direction of some biochemical reactions can reverse upon small perturbations of the surrounding conditions. For reactions with a small ∆*G*, the **same enzyme** can easily catalyze both the forward and reverse reactions under physiological conditions.

For the reaction shown in Figure 1, the passage states that the enzyme creatine kinase catalyzes **both the forward and reverse reactions**. Moreover, the passage states that the reaction flows in the **forward direction** under **physiological** anaerobic conditions and in the **reverse direction** under physiological aerobic conditions. Because physiological conditions allow the reaction to flow in both directions, and the reactions are catalyzed by the same enzyme (and not through a bypass reaction), **the reaction shown in Figure 1 has a small ∆*G* value near 0**.

Conversely, reactions that have a **large negative ∆*G*** are considered **spontaneous and irreversible** because the reverse reaction would have a correspondingly large but *positive* ∆*G*. For the reaction shown in Figure 2, the passage states that the cyclization reaction is **irreversible**. The reaction proceeds **spontaneously** and **only in the forward direction**. Therefore, the **reaction shown in Figure 2 has a large negative ∆*G***.

(Choices A and C) The reaction shown in Figure 1 is reversible and proceeds in the forward direction; therefore, its ∆*G* is near 0. It is neither large and negative (ie, irreversible) nor large and positive (ie, nonspontaneous and unlikely to occur at physiological conditions).

(Choice B) The reaction shown in Figure 2 proceeds irreversibly in the forward direction. Because the reaction is spontaneous in the forward direction, its ∆*G* value is negative, not positive.

Educational objective:
Reversible biochemical reactions have a small ∆*G* near 0 and can proceed in both the forward and reverse directions under physiological conditions. Biochemical reactions with a large negative ∆*G* are spontaneous but irreversible.

Time spent:0 sec
Subject:
Biochemistry

QID:404002
System:
Metabolic Reactions

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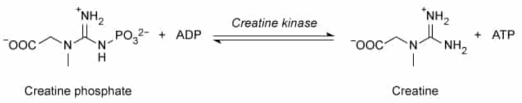

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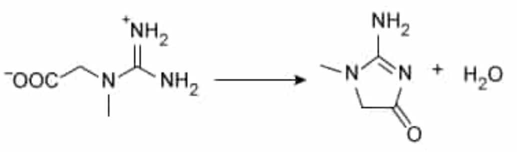

Figure 2 Production of creatinine from creatine

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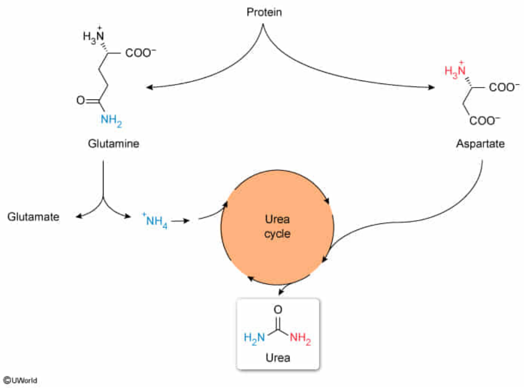
Most of the urea measured in BUN levels comes from the breakdown of:

- A. monosaccharides. [3%]
- B. purine nucleotides. [15%]
- ✔ C. protein. [78%]
- D. triglycerides. [3%]

Correct
77% answered correctly

Explanation:



Nitrogen in proteins is excreted as urea

The **catabolism** (ie, breakdown) of macromolecules and their monomers constitutes an important set of metabolic pathways to either **provide energy** or to **handle waste products** for a cell. The **carbon skeletons** of sugars, triglycerides, and proteins can typically feed into either the glycolysis or the aerobic respiration pathways to be oxidized into **carbon dioxide**, reducing oxygen to water and producing usable high-energy ATP.

However, in addition to carbon, the amino acids that serve as the building blocks of proteins all have **at least one amine group** and therefore contain **nitrogen atoms** that are *not* catabolized through glycolysis or the citric acid cycle. Instead, the nitrogen atoms of amino acids and proteins are integrated into the molecule urea, which is produced by the urea cycle.

The urea cycle occurs in the **liver** of animals and **spans both the cytosol and mitochondria**. Each molecule of urea produced contains **two nitrogen atoms**. The first nitrogen atom comes from a free ammonium ion (NH_4^+), which may be produced through deamidation of glutamine or asparagine side chains, or by deamination of any other amine (including the α-amine of α-amino acids). The second nitrogen atom in urea comes from the amino acid aspartate, although any amino acid can become aspartate by donating its α-amino group to oxaloacetate through transamination.

Although free ammonium in the liver can come from a variety of nitrogen-containing molecules, most free ammonium, as well as all of the nitrogen atoms that form urea's second $–NH_2$ group, comes from **amino acids and the breakdown of proteins**.

(Choice A) Most monosaccharides do not contain nitrogen atoms that would enter the urea cycle to produce urea. Although amino derivatives of monosaccharides may contribute to urea production, their contribution is small compared to proteins.

(Choice B) Although *pyrimidine* breakdown may contribute a small amount to the urea cycle, *purine* degradation results in the production of uric acid, not urea.

(Choice D) Triglycerides do not contain nitrogen atoms that would enter the urea cycle to produce urea. Although the head groups of *phospholipids* may contain nitrogen atoms, their contribution to urea production is small compared to proteins.

Educational objective:
Proteins and amino acids have nitrogen atoms that cannot be catabolized through glycolysis and the citric acid cycle. Instead, the nitrogen atoms are affixed to a molecule called urea, which is produced through the urea cycle.